Neptunium

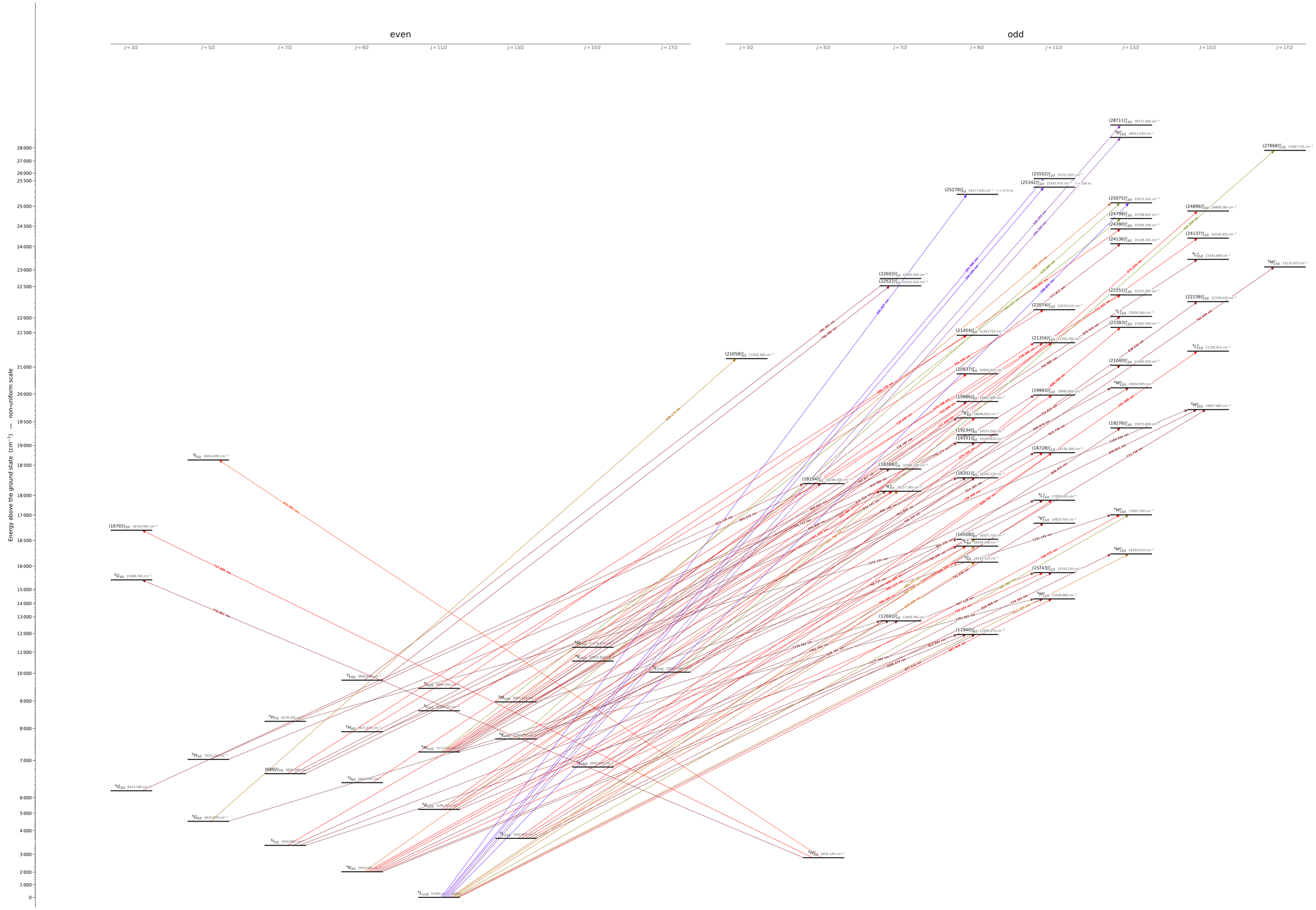
Np (Np I). Classified transitions below 2 µm (levels and lines from the literature): 77 levels, 90 lines.

Np⁺ ionisation limit 50535.54 cm^{-1} = 6.26561 eV (197.881 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient *A*. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	|⟨J‖er‖J'⟩| (ea₀)	A (s⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$^6L_{11/2}$ – (25552)$^\circ_{11/2}$	391.3599	391.2491	766.0274	–	–	–	–	first step of the strong one-colour two-photon ionization…
$^6L_{11/2}$ – (25342)$^\circ_{11/2}$	394.5945	394.4828	759.7483	–	–	0.692	–	first step of a Mainz two-step resonance-ionization scheme
$^6L_{11/2}$ – (25278)$^\circ_{9/2}$	395.6066	395.4947	757.8044	–	–	0.92	–	first step
$^6L_{11/2}$ – (25075)$^\circ_{13/2}$	398.8013	398.6885	751.7339	–	–	–	–	first step

λ, ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{237}Np hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$^6L_{11/2}$	776.08(20)	928(10)	F=3: -12227.44 F=4: -9406.59 F=5: -5766.62 F=6: -1231.63 F=7: +4289.52 F=8: +10903.10
$^6K_{9/2}$	861.300	-2158.500	F=2: -12832.19 F=3: -9600.74 F=4: -5543.96 F=5: -877.71 F=6: +4128.20 F=7: +9150.00
$^6L_{13/2}$	685.300	869.400	F=4: -12514.99 F=5: -9356.00 F=6: -5454.86 F=7: -751.39 F=8: +4824.64 F=9: +11353.48
$^4G_{5/2}$	1275.300	66.000	F=0: -11112.67 F=1: -9853.22 F=2: -7328.35 F=3: -3526.22 F=4: +1571.03 F=5: +7987.12
(6892)$_{7/2}$	892.800	-1783.800	F=1: -10999.61 F=2: -8704.35 F=3: -5452.59 F=4: -1473.66 F=5: +2926.63 F=6: +7366.05
$^6L_{15/2}$	617.600	980.300	F=5: -12766.89 F=6: -9363.78 F=7: -5266.05 F=8: -414.88 F=9: +5256.95 F=10: +11825.08
$^6H_{7/2}$	953.300	-1699.800	F=0: -9531.24 F=1: -8169.98 F=2: -5600.46 F=3: -2128.63 F=4: +1786.56 F=5: +5533.18
$^8M_{11/2}$	-926.700	-542.600	F=3: +14834.44 F=4: +11293.38 F=5: +6800.46 F=6: +1311.29 F=7: -5227.40 F=8: -12877.77
$^6K_{13/2}$	740.800	-1169.200	F=4: -14339.69 F=5: -10275.94 F=6: -5547.83 F=7: -236.32 F=8: +5564.17 F=9: +11745.70
$^4H_{9/2}$	793.000	-2251.400	F=2: -11935.64 F=3: -8881.22 F=4: -5071.32 F=5: -731.09 F=6: +3858.05 F=7: +8358.40
$^6H_{7/2}$	891.600	314.800	F=1: -9861.86 F=2: -8168.60 F=3: -5594.99 F=4: -2100.54 F=5: +2368.70 F=6: +7880.20
$^8L_{9/2}$	-1165.300	-698.500	F=2: +15702.73 F=3: +12416.38 F=4: +7953.09 F=5: +2243.00 F=6: -4801.18 F=7: -13284.25
$^6L_{17/2}$	584.000	1199.200	F=6: -13451.16 F=7: -9733.50 F=8: -5326.03 F=9: -165.26 F=10: +5820.23 F=11: +12709.80
$^6K_{15/2}$	686.800	-665.500	F=5: -14836.93 F=6: -10510.78 F=7: -5550.11 F=8: +5.13 F=9: +6109.32 F=10: +12711.12
$^8M_{15/2}$	1220.800	-167.900	F=5: -26003.16 F=6: -18626.55 F=7: -10042.34 F=8: -260.59 F=9: +10707.18 F=10: +22848.03
(11940)$^\circ_{9/2}$	1165.000	-611.600	F=2: -16299.07 F=3: -12620.59 F=4: -7787.30 F=5: -1860.37 F=6: +5083.76 F=7: +12953.35
(12693)$^\circ_{7/2}$	500.400	-806.400	F=1: -6061.50 F=2: -4830.30 F=3: -3069.90 F=4: -883.98 F=5: +1589.22 F=6: +4176.90
$^8M^\circ_{11/2}$	-644.600	2779.100	F=3: +11624.29 F=4: +8197.00 F=5: +4253.96 F=6: +22.55 F=7: -4224.37 F=8: -8168.48
(15743)$^\circ_{11/2}$	959.000	-626.600	F=3: -15842.93 F=4: -11815.54 F=5: -6858.19 F=6: -1022.16 F=7: +5631.02 F=8: +13029.60
$^4I^\circ_{9/2}$	585.500	-2014.600	F=2: -8973.98 F=3: -6613.10 F=4: -3700.30 F=5: -437.03 F=6: +2924.87 F=7: +6083.23
$^8M^\circ_{15/2}$	1440.500	1337.100	F=4: -26495.11 F=5: -19704.02 F=6: -11385.01 F=7: -1445.51 F=8: +10222.49 F=9: +23742.40
(16508)$^\circ_{9/2}$	298.600	-680.500	F=2: -4417.65 F=3: -3317.70 F=4: -1930.49 F=5: -324.07 F=6: +1416.49 F=7: +3189.12
$^6M^\circ_{13/2}$	989.300	1570.900	F=4: -17945.18 F=5: -13482.04 F=6: -7926.88 F=7: -1170.95 F=8: +6912.62 F=9: +16468.85
(18194)$^\circ_{9/2}$	852.900	-63.000	F=0: -7506.98 F=1: -6638.95 F=2: -4908.59 F=3: -2327.20 F=4: +1088.17 F=5: +5314.88
(18201)$^\circ_{9/2}$	260.800	-2985.900	F=2: -4954.54 F=3: -3276.37 F=4: -1387.16 F=5: +414.49 F=6: +1755.35 F=7: +2187.53
(18388)$^\circ_{7/2}$	376.200	-3252.700	F=1: -5974.77 F=2: -4293.02 F=3: -2118.91 F=4: +129.36 F=5: +1894.19 F=6: +2478.57
(18728)$^\circ_{11/2}$	1553.800	-818.400	F=3: -25587.77 F=4: -19122.59 F=5: -11141.55 F=6: -1711.61 F=7: +9086.87 F=8: +21160.15
(19101)$^\circ_{9/2}$	1433.300	-362.700	F=2: -19874.11 F=3: -15465.40 F=4: -9629.44 F=5: -2402.49 F=6: +6170.11 F=7: +16033.95
(19234)$^\circ_{9/2}$	1693.800	-1352.100	F=2: -23909.46 F=3: -18422.43 F=4: -11264.14 F=5: -2569.79 F=6: +7491.60 F=7: +18717.22
(19276)$^\circ_{13/2}$	889.800	-377.700	F=4: -16829.02 F=5: -12263.80 F=6: -6833.48 F=7: -564.21 F=8: +6513.52 F=9: +14364.83
$^8K^\circ_{9/2}$	1061.300	-2437.300	F=2: -15709.97 F=3: -11794.88 F=4: -6859.11 F=5: -1146.40 F=6: +5038.61 F=7: +11330.30
$^8M^\circ_{15/2}$	1729.500	1265.100	F=5: -36291.02 F=6: -26304.39 F=7: -14488.86 F=8: -768.53 F=9: +14943.36 F=10: +32744.40
(19986)$^\circ_{9/2}$	616.100	-1981.600	F=2: -9379.61 F=3: -6936.83 F=4: -3910.97 F=5: -500.21 F=6: +3047.77 F=7: +6435.73
$^6M^\circ_{13/2}$	722.800	797.400	F=4: -13245.81 F=5: -9877.16 F=6: -5733.58 F=7: -759.85 F=8: +5108.42 F=9: +11944.85
(20837)$^\circ_{9/2}$	584.300	-548.600	F=2: -8285.57 F=3: -6368.09 F=4: -3875.45 F=5: -862.52 F=6: +2602.14 F=7: +6436.23
(21040)$^\circ_{13/2}$	1561.000	1534.900	F=4: -28678.40 F=5: -21345.68 F=6: -12351.60 F=7: -1589.90 F=8: +11063.40 F=9: +25749.97
(21593)$^\circ_{13/2}$	1111.600	-1067.300	F=4: -21253.00 F=5: -15366.60 F=6: -8438.39 F=7: -542.25 F=8: +8235.61 F=9: +17796.67
$^4L^\circ_{13/2}$	885.300	-539.600	F=4: -16806.91 F=5: -12214.38 F=6: -6771.83 F=7: -516.62 F=8: +6507.67 F=9: +14251.23
(22074)$^\circ_{11/2}$	-328.900	-1250.100	F=3: +4827.54 F=4: +3893.79 F=5: +2573.18 F=6: +763.43 F=7: -1658.20 F=8: -4834.90
(22238)$^\circ_{15/2}$	828.000	-1091.200	F=5: -17992.51 F=6: -12687.80 F=7: -6640.82 F=8: +82.95 F=9: +7408.68 F=10: +15252.20
(22251)$^\circ_{13/2}$	1252.500	-815.400	F=4: -23797.99 F=5: -17284.60 F=6: -9572.02 F=7: -716.71 F=8: +9215.48 F=9: +20149.28
(22522)$^\circ_{7/2}$	-48.000	659.500	F=1: +893.30 F=2: +608.88 F=3: +252.89 F=4: -89.85 F=5: -306.30 F=6: -255.12
$^8M^\circ_{17/2}$	1660.900	1046.300	F=6: -39080.94 F=7: -27777.76 F=8: -14721.36 F=9: +143.65 F=10: +16879.59 F=11: +35555.70
$^8L^\circ_{15/2}$	1165.300	1594.900	F=5: -24181.63 F=6: -17681.97 F=7: -9891.69 F=8: -715.11 F=9: +9957.14 F=10: +22248.10
(24137)$^\circ_{15/2}$	1573.900	1280.100	F=5: -32979.05 F=6: -23930.65 F=7: -13207.78 F=8: -733.62 F=9: +13579.61 F=10: +29830.65
(24896)$^\circ_{15/2}$	1423.700	-1642.900	F=5: -30852.11 F=6: -21802.96 F=7: -11459.19 F=8: +80.62 F=9: +12703.81 F=10: +26283.65
(25075)$^\circ_{13/2}$	1470.09(18)	323(8)	F=4: -27439.96 F=5: -20188.89 F=6: -11446.62 F=7: -1190.77 F=8: +10604.73 F=9: +23969.71
(25278)$^\circ_{9/2}$	570.08(22)	-302(9)	F=2: -7977.02 F=3: -6176.18 F=4: -3810.29 F=5: -909.56 F=6: +2488.27 F=7: +6337.90

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^6L_{11/2}$ – (25278)$^\circ_{9/2}$	395.607	239-237	-7892(13)
$^6L_{11/2}$ – (25075)$^\circ_{13/2}$	398.801	239-237	-8168(17)

Shift = ν(first isotope) − ν(second isotope).